
DESCRIPTIVE COMPANION NOTE

Cash, Inflation, and Household Optionality

A Public-Data Companion Note on Liquidity, Purchasing Power, and
Emergency Buffers

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Abstract

Cash-like household liquidity has two margins that should not be collapsed into a slogan. On one margin, idle nominal balances lose purchasing power when inflation exceeds yield. On another, liquid balances can preserve household optionality by reducing the chance that ordinary expense or income shocks turn into high-cost borrowing, arrears, skipped care, forced sales, or interrupted long-horizon plans. This revised companion note narrows the claim accordingly. It does not estimate the marginal option value of liquidity and does not claim to be a journal-style research contribution. It uses public FRED data and Federal Reserve SHED public tables to document the accounting tradeoff: 3-month Treasury bill benchmark returns were negative in exact ex-post real terms during 2021–2022 but positive by 2023–2025, while zero-yield nominal balances continued to lose real value when prices rose. SHED 2025 public tables show that 37 percent of adults would not cover a \$400 emergency expense using cash or its equivalent, with higher rates among young adults, parents, adults with less education, Black adults, and Hispanic adults. The paper's contribution is therefore interpretive and measurement-oriented: distinguish Treasury bill benchmark yield, zero-yield nominal balances, and cash-equivalent emergency-payment capacity; explain why liquidity has an option-like role; and identify the empirical work needed to estimate that option value directly.

1. Motivation and research gap

A household liquidity framework needs to hold two facts at once. First, idle nominal balances are inflation-exposed. A dollar kept in a zero-yield account buys less when consumer prices rise. Second, liquid balances can protect household choice. A household with accessible liquidity is less likely to turn a car repair, medical bill, rent timing mismatch, insurance deductible, missed shift, or family obligation into expensive debt, missed payments, delayed care, or forced liquidation.

The research gap is not whether these facts are individually true. Both are well understood. The useful question is how to connect them: when does the household value of liquidity exceed the inflation/yield opportunity cost of holding it?

This note does not fully answer that question. It is a descriptive companion note that sets up the measurement problem. The stronger research paper would estimate the marginal value of liquid wealth using household microdata and observed hardship outcomes after comparable shocks. The

present contribution is narrower: define the margins cleanly, fix the language around “cash,” and show the public-data facts that motivate deeper work.

2. Measurement setup and definitions

The note separates three objects that are often conflated.

3-month Treasury bill benchmark yield is the short safe-rate proxy used for the return side of the note. The FRED series is `TB3MS`, a monthly, not-seasonally-adjusted, discount-basis 3-month Treasury bill secondary-market rate. It is not a checking-account return, deposit-account return, money-market-fund total return, or realized household portfolio return.

Zero-yield nominal balances are fixed-dollar balances earning no interest. This object is useful because transaction balances often sit in low-yield form. It is not a claim that all households hold all liquidity at zero yield.

Cash-equivalent emergency-payment capacity is the SHED public-table concept for whether adults would cover a hypothetical \$400 emergency expense using cash, savings, or a credit card paid off at the next statement. It is not literal cash in a bank account, and “would not cover using cash or equivalent” is not the same as “cannot pay by any method.”

The real-return calculation uses annual average `TB3MS` and December-to-December CPI inflation from `CPIAUCSL`. The exact ex-post real benchmark is:

$$\text{real return} = [(1 + \text{nominal yield}) / (1 + \text{inflation})] - 1.$$

For transparency, Table 1 also reports the simpler approximation, nominal yield minus inflation. The approximation is close in ordinary years but less exact in high-inflation periods. `CPIAUCSL` is seasonally adjusted CPI-U; a stricter archival version should document vintage and access date.

3. A minimal optionality framework

A simple two-period household model clarifies the tradeoff. In period 1, a household chooses liquid wealth `L`. The liquid asset earns real return `r_L`, which may be low or negative after inflation. An illiquid or long-horizon alternative earns expected real return `r_A`, but cannot be used costlessly for emergency payment. In period 2, the household faces a shock `S` with probability `p`. If liquid wealth is insufficient, the household pays an adjustment cost `C`, such as high-cost borrowing, arrears fees, missed care, forced sale, or disrupted plans.

The marginal value of one more dollar of liquidity can be written conceptually as:

$$\text{MVL} = \text{expected avoided adjustment cost} + \text{consumption-smoothing value} - \text{opportunity cost}.$$

The opportunity cost is roughly the return gap between the best alternative use of the dollar and the liquid instrument. The avoided adjustment cost depends on shock probability, credit access, penalty costs, dependents, insurance, job stability, and informal support.

This model yields straightforward comparative statics. Liquidity should be more valuable for households with higher shock risk, weaker credit access, higher arrears/default penalties, more binding transportation or medical needs, more dependents, or lower insurance coverage. Excess liquidity beyond likely emergency needs should have lower marginal option value and may become mostly inflation/yield drag.

The model also explains why the descriptive SHED table is not enough. Observing that households differ in emergency-payment capacity does not estimate **MVL**. Estimating **MVL** requires linking pre-shock liquidity to post-shock outcomes among otherwise comparable households.

4. The real-return side of short liquidity

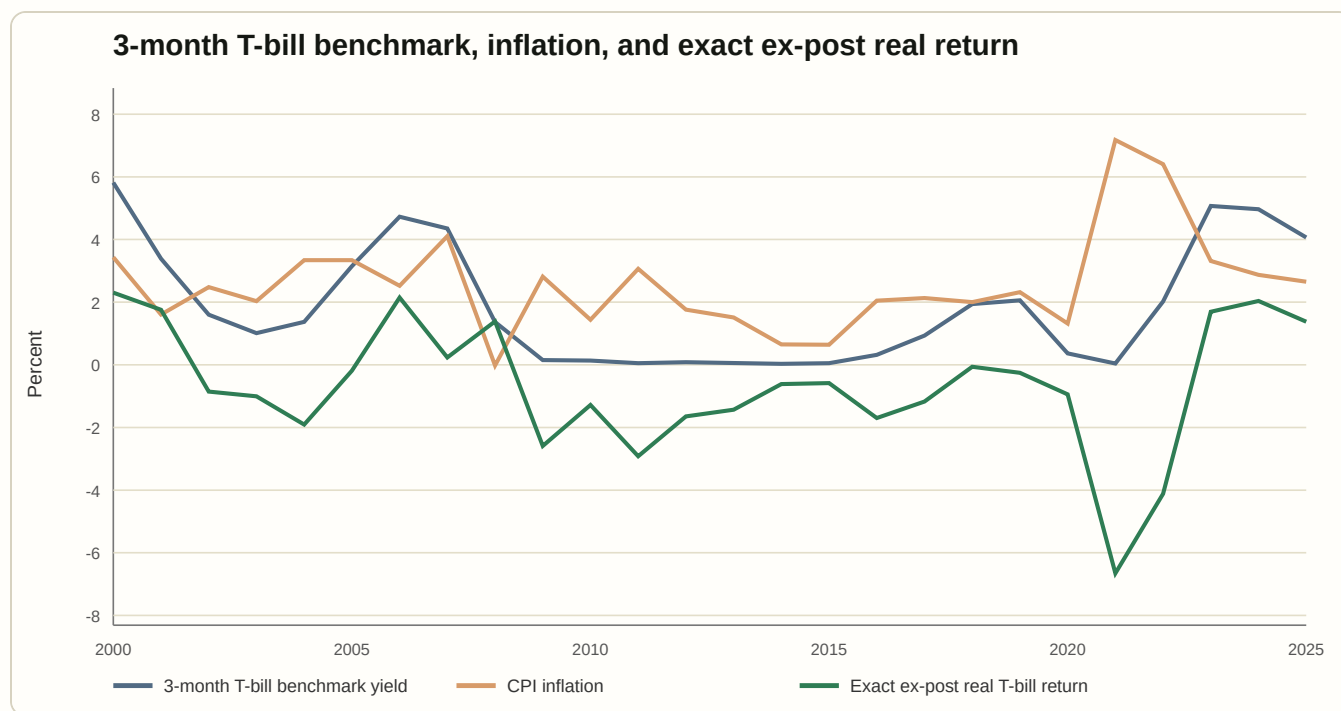


Figure 1. 3-month Treasury bill benchmark yield, CPI inflation, and exact ex-post real T-bill benchmark return.

Figure 1 shows the short-rate and inflation side of the tradeoff. During 2021 and 2022, the 3-month Treasury bill benchmark return was negative in exact ex-post real terms. By 2023–2025, short yields exceeded CPI inflation in this annual screen.

The figure should not be read as a household deposit-return series. Many households do not receive the Treasury bill benchmark on transaction balances. The wedge between available short safe yields and realized household deposit yields is itself a useful research object.

Table 1. Recent short-liquidity benchmark and inflation context.

Year	T-bill yield	CPI inflation	Exact ex-post real T-bill	Approx real T-bill	Zero-yield real return
2019	2.1	2.3	-0.3	-0.3	-2.3
2020	0.4	1.3	-0.9	-1.0	-1.3
2021	0.0	7.2	-6.7	-7.1	-6.7
2022	2.0	6.4	-4.1	-4.4	-6.0
2023	5.1	3.3	1.7	1.8	-3.2
2024	5.0	2.9	2.0	2.1	-2.8
2025	4.1	2.7	1.4	1.4	-2.6

Source note: FRED `TB3MS` and `CPIAUCSL` ; annual average T-bill benchmark yield; December-to-December CPI inflation; exact real return computed as $[(1 + \text{yield}) / (1 + \text{inflation})] - 1$; approximation equals yield minus inflation.

5. Zero-yield nominal balances

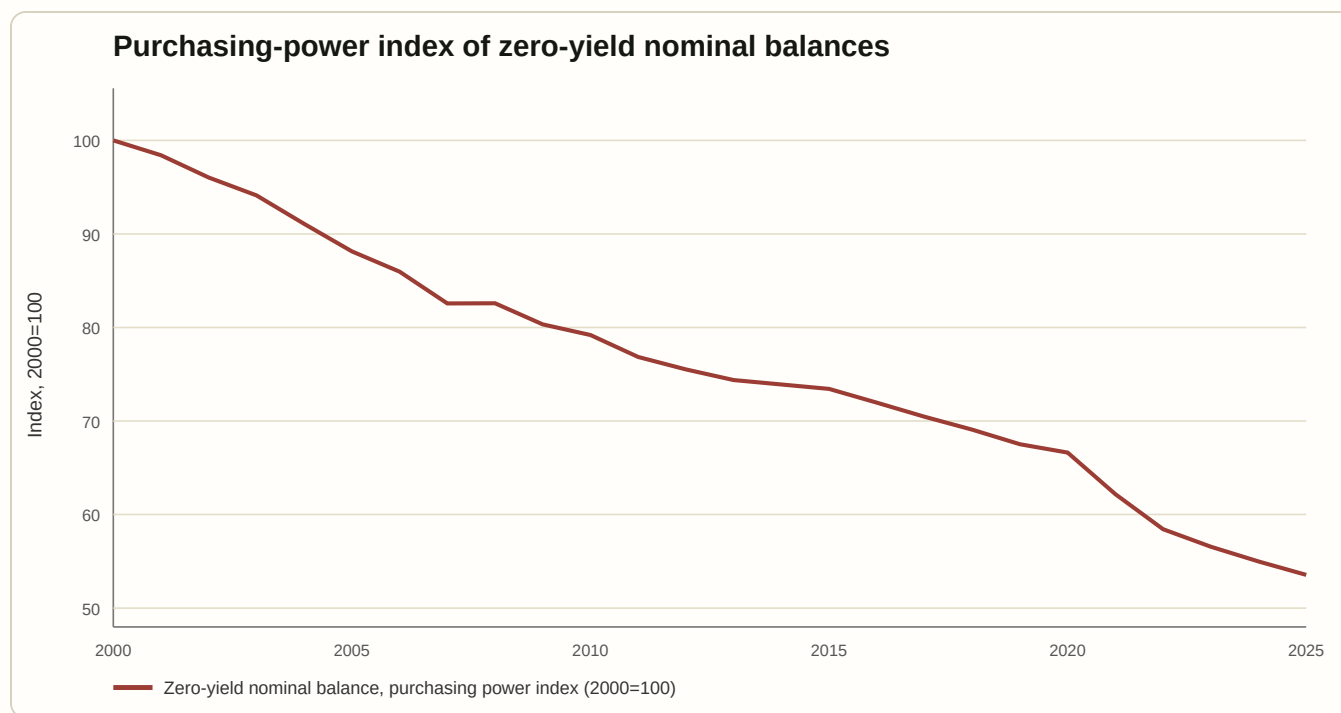


Figure 2. Purchasing-power index of a fixed zero-yield nominal balance, normalized to 100 in 2000.

Figure 2 isolates a different object: a fixed nominal balance earning no yield. An indexed chart avoids false precision about the dollar value of a specific household balance. It simply shows that a zero-yield nominal dollar loses real purchasing power as consumer prices rise.

This is the part of the cash critique that is correct. Excess idle balances can impose a real carrying cost under inflation. But that statement alone does not determine the right emergency buffer, because the return calculation ignores the option value of avoiding forced adjustment after shocks.

6. Emergency-payment optionality in SHED public tables

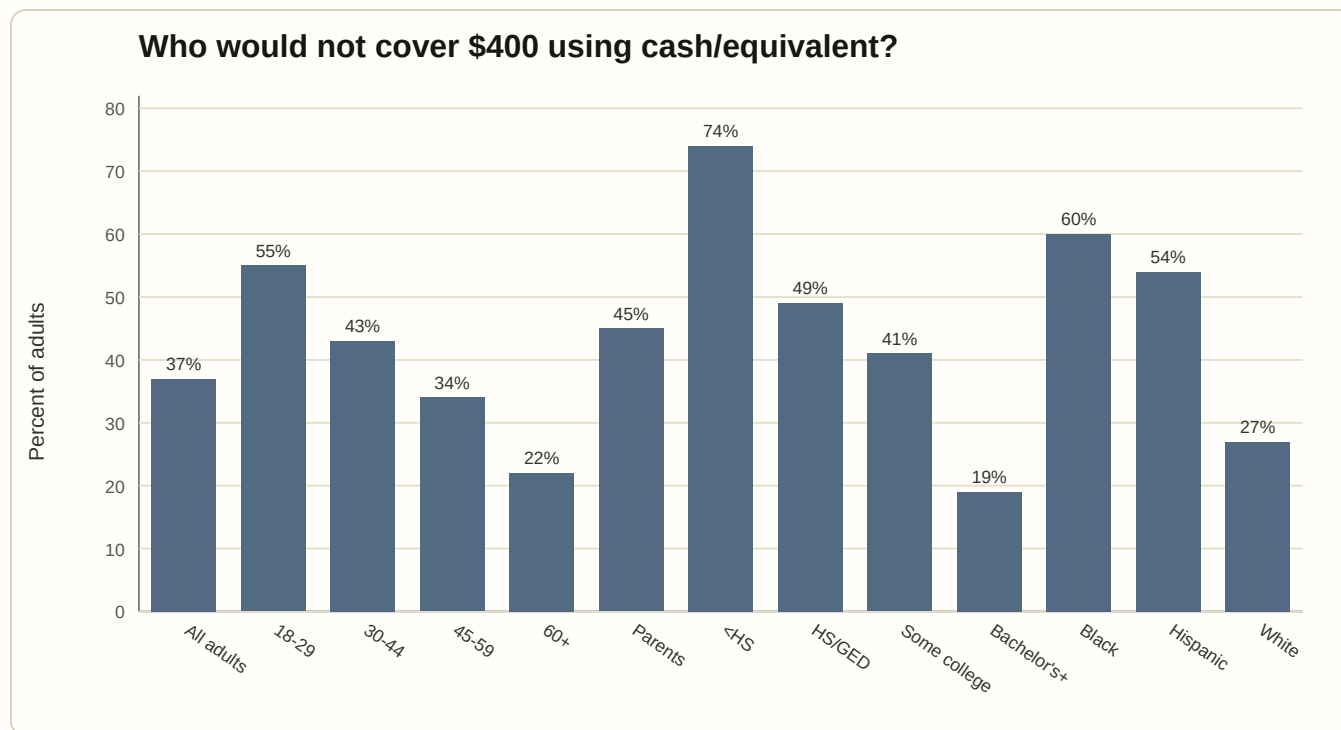


Figure 3. SHED 2025 share of adults who would not cover a \$400 emergency expense using cash or its equivalent.

SHED public tables provide the note's clearest descriptive evidence. In 2025, 63 percent of adults reported that they would cover a \$400 emergency expense using cash or its equivalent, implying that 37 percent would not. The Federal Reserve describes the 2025 survey as based on nearly 13,000 adults surveyed in October 2025 and released in May 2026.

Table 2. SHED 2025 cash-equivalent emergency-payment capacity.

Group	Can cover \$400 cash/equiv.	Would not cover using cash/equiv.
All adults	63.0	37.0
18-29	45.0	55.0
30-44	57.0	43.0
45-59	66.0	34.0
60+	78.0	22.0
Parents	55.0	45.0
<HS	26.0	74.0
HS/GED	51.0	49.0
Some college	59.0	41.0
Bachelor's+	81.0	19.0
Black	40.0	60.0
Hispanic	46.0	54.0
White	73.0	27.0

Source note: Federal Reserve SHED public tables for the 2025 survey. “Cash/equivalent” includes cash, savings, or a credit card paid off at the next statement. Public percentages are rounded survey estimates; this note does not compute confidence intervals from microdata.

The distributional pattern is important but not causal. In the public tables, 55 percent of adults ages 18–29, 45 percent of parents with own children under 18, 74 percent of adults with less than a high school education, 60 percent of Black adults, and 54 percent of Hispanic adults would not cover the \$400 expense using cash or equivalent.

This should not be restated as inability to pay by any means. Some households would use another method. The narrower interpretation is still economically meaningful: lacking cash-equivalent payment capacity suggests a higher chance that a small emergency becomes borrowing, payment delay, or another adjustment.

7. What this note can and cannot claim

The note supports four descriptive claims.

First, zero-yield nominal balances are inflation-exposed.

Second, the 3-month Treasury bill benchmark is not the same as household deposit yield. When the benchmark exceeds inflation, households still may not receive that return on ordinary transaction balances.

Third, cash-equivalent emergency-payment capacity is unevenly distributed. SHED public tables show large gaps across age, parental status, education, and race/ethnicity.

Fourth, emergency liquidity can have option value even when its observed yield is low. The value depends on avoided costly adjustment, not only on asset return.

The note does not estimate an optimal cash buffer. It does not estimate the causal effect of liquidity on hardship. It does not model taxes, account access, account fees, behavioral frictions, deposit-yield pass-through, or household-specific risk. It does not compare liquidity with any specific long-duration asset. It also does not treat the \$400 threshold as a complete measure of emergency preparedness.

8. Planner relevance

The practical implication is a sequencing principle, not an allocation rule. Emergency liquidity can protect the ability to maintain long-horizon plans. Without it, households may be forced to sell assets, carry high-interest debt, miss payments, or pause accumulation after predictable shocks.

A household-resilience planner should therefore separate four buckets:

1. transaction balances for near-term bills;
2. emergency liquidity for unexpected shocks;
3. planned-spending liquidity for known near-term expenses;
4. excess idle balances that may be exposed to avoidable inflation/yield drag.

The key distinction is between liquidity that protects optionality and idle balances that sit beyond practical buffer needs.

9. Next empirical step

The journal-style version of this project would be a different paper. It would estimate whether households with emergency liquidity are less likely to experience hardship after comparable shocks.

A credible empirical design would require household microdata from SHED, SCF, PSID, Consumer Expenditure Survey, administrative bank-account data, or another source that observes pre-shock liquidity and post-shock hardship. It would define shocks explicitly: job loss, hours reduction, medical expense, car repair, rent increase, insurance deductible, family transfer obligation, or income volatility.

The outcomes should include arrears, high-cost borrowing, skipped care, credit-card balance carrying, forced asset sales, interrupted saving, and disrupted household plans. The analysis should compare simple liquidity measures with richer categories: checking, savings, money-market funds, Treasury bills, credit-card float, and informal support.

The welfare object would be the expected avoided hardship cost from an additional dollar of liquidity compared with the after-tax, inflation-adjusted opportunity cost of holding that dollar in liquid form.

10. Conclusion

Cash-like liquidity is both inflation-exposed purchasing power and household optionality. Treating it as only one or the other produces bad reasoning. The public-data facts are simple: idle nominal balances lose real value when prices rise; the short safe-rate benchmark can be negative or positive in real terms depending on the inflation and rate environment; and many households would not cover a modest emergency expense using cash or equivalent.

The note remains descriptive. Its contribution is to clarify the measurement objects and frame the next empirical task. The stronger research question is not whether cash has costs or benefits. It is how large the marginal value of emergency liquidity is for different households, and whether that value exceeds the inflation/yield opportunity cost of holding liquid balances.

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